

Design and Development of an Intelligent Operation Center (IOC) for Smart Building Fire Detection and Automated Control

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Abstract—Urban and building fires pose a significant threat to life and infrastructure, demanding rapid and reliable detection systems. Traditional fire monitoring heavily relies on physical sensors, which often suffer from delayed response times, while purely vision-based Artificial Intelligence (AI) systems are prone to false positives in complex environments. In this paper, we propose a hybrid Intelligent Operation Center (IOC) for building safety that synergizes an Internet of Things (IoT) sensor network with Edge AI computer vision. Our architecture leverages an advanced NVIDIA Jetson Orin Nano to run real-time visual flame and smoke identification. While existing literature primarily focuses on YOLOv8 architectures, our empirical tests revealed vulnerabilities to artificial indoor lighting. Consequently, we upgraded our visual module to a highly optimized YOLO11-medium model. This visual data is continuously fused with telemetry from physical sensors (temperature and MQ-2 gas detectors) transmitted via a local Wi-Fi network. The multimodal data stream is centralized and managed through a real-time React-based dashboard utilizing a WebSocket relay server, enabling automated emergency responses. This edge-based, sensor-fusion approach significantly reduces false alarms, eliminates cloud-dependency latency, and provides facility managers with a robust decision support system for critical emergencies.

Index Terms—Intelligent Operation Center (IOC), Edge AI, Fire Detection, Sensor Fusion, Wi-Fi, NVIDIA Jetson Orin Nano, YOLOv8, YOLO11.

I. INTRODUCTION

Urban and building fires pose a significant threat to human life, infrastructure, and the environment. Traditional fire monitoring systems heavily rely on physical sensors, such as smoke (MQ-2) and temperature detectors. While effective, these conventional systems require the physical presence of smoke or heat to reach the sensor before an alarm is triggered, which can result in crucial delays during the initial phases of a fire. Furthermore, physical sensors in complex environments can be prone to false alarms or failure due to localized damage.

To overcome these limitations, integrating Computer Vision (CV) and Artificial Intelligence (AI) into fire safety systems has become a highly researched domain. By analyzing real-time video feeds, AI models can detect the visual signatures of flames and smoke instantly, often before physical parameters trigger traditional sensors. However, deploying deep learning models for real-time video processing requires significant computational power, which is challenging in building

management systems that rely on low-cost, low-power edge devices.

In this paper, we propose an Intelligent Operation Center (IOC) for building safety that synergizes a classic Internet of Things (IoT) sensor network (communicating via Wi-Fi) with an Edge AI computer vision module. Specifically, we utilize an NVIDIA Jetson Orin Nano paired with an IP camera to run lightweight object detection models for real-time flame and smoke identification. Starting from a baseline YOLOv8 model inspired by current literature, we evolved our vision pipeline to a YOLO11 architecture to specifically address and mitigate false positives caused by indoor artificial lighting.

The data from the visual AI module is fused with the telemetry from physical sensors (temperature, gas/smoke). Instead of relying on high-latency cloud databases, this unified data stream is managed locally via a custom WebSocket relay server and monitored through a real-time React-based dashboard. This enables instantaneous, automated emergency responses, such as triggering physical alarm bells, warning lights, and alerting building managers.

The main scientific contributions of this work are:

- The design and implementation of a hybrid fire detection architecture combining Wi-Fi-based physical sensors with Edge AI vision processing.
- The deployment and evaluation of a lightweight deep learning model optimized for the resource-constrained NVIDIA Jetson Orin Nano, balancing detection accuracy with real-time inference speed.
- A comprehensive dashboard integration that aggregates multimodal data (visual and physical) to reduce false positives and improve decision-making in critical situations.

The remainder of this paper is organized as follows: Section II provides an overview of related work in computer vision for fire detection. Section III details the methodology, encompassing the overall system architecture, the edge AI implementation, and IoT sensor fusion. Section IV is dedicated to the experimental setup and dataset usage. Results and discussions are detailed in Section V. Lastly, Section VI serves as the conclusion.

II. RELATED WORK

The design of an Intelligent Operation Center (IOC) for building fire safety and energy management draws upon a broad spectrum of research domains, ranging from vision-based fire and smoke detection to Building Energy Management Systems (BEMS) and smart city operation centers.

This section reviews and compares the most relevant studies, organized into four thematic pillars: (i) traditional and sensor-based fire detection methods, (ii) deep learning models for visual fire and smoke detection, (iii) edge AI deployment for real-time inference, and (iv) intelligent building and operation center architectures.

A. Traditional and Sensor-Based Fire Detection

Conventional fire detection systems rely primarily on physical sensors that measure heat, smoke concentration, or ambient gas levels. A comprehensive survey of fire sensing technologies carried out by Ravikumar and Pau [2] analyses the key signatures of a fire event—flame, smoke, temperature, and volatile organic compounds (VOCs)—and evaluates the performance of the corresponding sensor families in terms of detection latency, false alarm rate, and spatial coverage. The review concludes that, although physical sensors offer low unit cost and straightforward installation, they suffer from high false alarm rates caused by environmental noise such as cooking fumes, steam, or dust, and their spatial coverage remains inherently limited to the immediate vicinity of the sensing element.

An alternative sensing paradigm is proposed by Sun et al. [12], who integrate a four-channel metal-oxide semiconductor (MOS) electronic nose (E-nose) into a household oven to detect burnt odors across multiple cooking stages. By combining a window-based feature extraction strategy with an Extreme Gradient Boosting (XGBoost) classifier, the system achieves an average real-time recognition accuracy of 91.3% on an embedded microcontroller without any PC dependency. While this approach is highly relevant for indoor kitchen environments, its generalization to broader building fire scenarios remains limited, as it is specifically tuned to oven emissions and a restricted set of food recipes.

A multi-sensor IoT hybrid model that addresses both indoor and outdoor scenarios is presented by Burnias [11]. A Raspberry Pi continuously monitors temperature, gas, smoke, and humidity readings; anomalies are flagged using Z-score analysis and forwarded over MQTT to an AWS cloud backend, where a hybrid model combining Fuzzy Logic and Random Forest Classifier (RFC) achieves approximately 99% accuracy on sensor data. A secondary ResNet50-based CNN model is then triggered to classify camera images into fire or no-fire categories with 90% accuracy, and the entire pipeline updates a Next.js web dashboard in under 774 ms. Despite its promising multi-modal design, this approach relies on cloud round-trips for inference, introducing latency and single-point-of-failure risks that are incompatible with the strict real-time requirements of a building fire safety system.

B. Deep Learning Models for Visual Fire and Smoke Detection

The limitations of purely sensor-based approaches have driven significant research activity toward camera-based solutions powered by deep learning. Elhanashi et al. [1] present a comprehensive survey of deep learning techniques applied to fire and smoke detection, covering CNNs, object detection

frameworks such as YOLO and Faster R-CNN, and spatiotemporal models for video analysis. The survey highlights important progress in detection accuracy and robustness, while also identifying persistent challenges: the scarcity and limited diversity of annotated training datasets, susceptibility to false alarms triggered by fog, strong sunlight, or reflections, and the high computational cost of deploying large models on resource-constrained platforms.

To address the dataset scarcity issue, Wang et al. [3] introduce the Flame And Smoke Detection Dataset (FASDD), an open benchmark comprising more than 120,000 heterogeneous images collected from ground-based, airborne, and space-borne sensors. Training Swin Transformer models on FASDD achieves mAP50 scores of 84.9%, 89.7%, and 74.0% on the three sensor-domain sub-datasets, and demonstrates effective wildfire localization from satellite imagery. Catargiu et al. [4] complement this effort by releasing a FireAndSmoke open dataset of over 22,000 images compiled from 1,200 YouTube videos, covering day/night, indoor/outdoor, and urban/forest scenarios. Their comparative evaluation of the YOLO detector family identifies YOLOv5 and YOLOv10 as the most suitable models for high-accuracy fire and smoke detection, with YOLOv10 reaching 89% precision and a mAP@50 above 91%.

Several works have proposed architectural improvements to standard YOLO models to better handle the irregular morphology of fire and smoke. Yu et al. [5] replace the default convolutional module of YOLOv5 with a lightweight self-designed block and add an ECA attention mechanism, achieving a precision of 91.22% and a recall of 93.78% on fire detection benchmarks. Zhang et al. [6] introduce the YOLOv8-FEP model, which augments the YOLOv8 backbone with a large-target detection head, an Efficient Multi-Scale Attention (EMA) module, and a PAN-Bag path aggregation structure; the resulting architecture improves mAP by 3.1% and accuracy by 5.8% over the baseline YOLOv8 model. Shen et al. [7] propose FireViT, a lightweight backbone that combines deformable convolution with a Vision Transformer (ViT) to capture the irregular spatial variations of smoke and flame; FireViT achieves 92.1% mAP on a natural-light fire dataset and 95.1% on an infrared fire dataset at only 12.5 GFLOPs.

Beyond purely frame-level detection, Gagliardi and Saponara [9] propose AdViSED, a video-based smoke detection algorithm for CCTV surveillance systems that exploits Kalman motion estimation, color analysis, blob labeling, and a M-of-N decisor. The method generates an alarm within a few seconds—15 times faster than the 60-second threshold prescribed by the EN50155 standard for onboard train fire-alarm systems—and reduces computational complexity by 63% compared to state-of-the-art techniques on the same hardware. Yadamakanti et al. [8] propose a hybrid approach that combines CNN-based context classification with HSV color segmentation and a temporal persistence module to eliminate false alarms caused by lights and reflections; tested on more than 500 experiments, the system detects fire in every positive sequence with no missed detection.

A sensor-vision fusion framework is evaluated by Dibra and

Gkonis [10], who combine IoT sensor readings (temperature, humidity, gas) with multiple YOLO nano variants within a decision-level explainable sensor fusion (ESF) framework. SHAP explanations for ML models and LIME explanations for YOLO detectors are integrated to improve interpretability. The ESF approach effectively mitigates the individual failure modes of vision-only systems (e.g., visual clutter) and sensor-only systems (e.g., limited spatial coverage), achieving perfect classification scores under controlled experimental conditions.

C. Edge AI Deployment for Real-Time Inference

Deploying deep learning models for fire detection on embedded edge platforms is a prerequisite for low-latency, infrastructure-independent operation. Aramendia et al. [13] benchmark the YOLOv5n model across three low-power platforms—NXP i.MX93, Xilinx Kria KV260, and NVIDIA Jetson Orin Nano—for fire and smoke detection in tunnels. The NVIDIA Jetson Orin Nano achieves the best detection accuracy and the fastest inference time, with an average execution time of 3.1 ms, making it the most suitable platform for high-performance real-time applications. The NXP i.MX93 consumes only 2.14 W in active mode, making it preferable for battery-powered deployments, while the Kria KV260 exhibits the highest power draw at 8.74 W.

Yang et al. [14] combine RGB and infrared imaging from FLIR thermal cameras with edge computing to build an automated fire detection and alarm system for urban buildings. Edge processing minimizes detection latency and enables faster alarm activation; cloud connectivity provides remote monitoring, scalable data storage, and a dashboard for authorized users. The system is validated on a university campus testbed and demonstrates reliable fire detection capability in risk-prone areas. Integrating both edge inference and cloud management in a single architecture aligns closely with the design philosophy of the IOC proposed in this paper.

D. Building Energy Management and Intelligent Operation Centers

Building Energy Management Systems (BEMS) form the backbone of energy efficiency and automated control in modern buildings. Jones et al. [15] present the hardware, software, and control methodology of a bespoke BEMS designed for a low-temperature community heat network connecting seven properties within the SCENIC project. The system uses a Raspberry Pi-based open-source platform (oBeMS) and demonstrates that a simple, low-cost hardware configuration can support flexible energy scheduling and monitoring. However, the authors explicitly note the absence of machine learning or predictive algorithms, identifying this as a direction for future work.

Hwang et al. [16] address this limitation by proposing DT-BEMS, which integrates Digital Twin technology with a BEMS. A virtual building model comprising an energy imputation sub-model, a Predicted Mean Vote (PMV) thermal comfort model, and an occupancy model is encoded using the Brick schema ontology. A rule-based Holistic Operational Signature (HOS) classifier identifies energy waste patterns

in real time; validated over one month of office data, DT-BEMS achieves 97.9% operational signature classification accuracy and identifies 23.5% of operations as inefficient, with unoccupied-period waste being the dominant category (19.9%).

Hannan et al. [17] review Internet of Energy (IoE)-based BEMS technologies for nearly or net zero energy buildings (nZEBs). The review examines energy router architectures, storage systems, renewable source integration, and plug-and-play interfaces, and identifies key open challenges: data reliability, real-time optimization, scalable IoE architectures, and the need for advanced AI-based controllers capable of handling the stochastic nature of renewable generation and occupant behavior.

At a larger scale, Zhuhadar et al. [18] analyse the architectural design of IBM's Intelligent Operation Center (IOC) as a "system of systems" that aggregates heterogeneous data streams from city infrastructure to support operational decision-making. The study demonstrates how the IOC concept—originally conceived for smart city management—can be transposed to university campuses, where it serves to unify multiple departments under a single monitoring and control umbrella, improving quality of service and resource efficiency.

The application of AI to city operation centers is further explored by Pratama and Suswanta [19], who document the Jambi City Operation Center (JCOC) in Indonesia. The JCOC integrates 15 applications and websites under a centralized 24-hour monitoring platform, enabling real-time response to citizen needs. Key lessons include the importance of inter-agency data integration and the critical role of human resource training in ensuring effective technology adoption. The study also highlights that many smart-city applications remain underutilized when ICT infrastructure or operator readiness is insufficient, a finding directly relevant to the deployment of the IOC described in this paper.

Khayat et al. [20] examine the analogous concept of the Security Operation Center (SOC) in the cybersecurity domain, proposing a reference architecture that integrates AI and ML for automated incident response, behavioral analytics, and threat prediction. While the SOC context differs from building fire safety, the architectural principles—continuous multi-source data ingestion, real-time AI-driven anomaly detection, and a unified dashboard for human operators—translate directly to the IOC paradigm.

Finally, Khan and Rafi [21] provide a foundational framework for Intelligent Decision Support Systems (IDSS), showing how knowledge management components that combine machine learning, case-based reasoning, and domain modeling can be integrated into decision-support workflows. This framework informs the design of the AI-assisted alarm and recommendation module of our IOC.

E. Research Gap and Proposed Contribution

Table I summarises the advantages and limitations of the surveyed works. Several gaps emerge from this review. First, existing sensor-based fire detection systems suffer from high false alarm rates and limited spatial coverage. Second, while

TABLE I
COMPARISON OF FIRE DETECTION APPROACHES

Criteria	Sensors	DL Models	Edge AI	IOC/BEMS
Detection type	Gas, heat, smoke	Visual (camera)	Visual (camera)	Multi-source
False alarm rate	High	Medium	Low	Low
Response time	Fast	Fast	Real-time	Near real-time
Edge deployment	Yes	No	Yes	Partial
Coverage	Local	Scene-level	Scene-level	Building-wide
Dashboard / IOC	No	No	No	Yes
AI integration	None	High	High	Medium
Energy mgmt.	No	No	No	Yes
This work	YOLO on Jetson Orin Nano + React IOC dashboard			

deep learning models (such as YOLOv5 to YOLOv10) achieve high detection accuracy, their deployment on real-time edge hardware remains under-explored, and the specific transition to the latest YOLO11 architecture to mitigate complex indoor reflections is absent from current literature. Third, no study to date integrates such advanced models within a building IOC that simultaneously manages real-time IoT sensor telemetry (without cloud latency) and visual fire detection.

The system proposed in this paper addresses these gaps. We demonstrate the progression from a baseline YOLOv8 model to a highly robust YOLO11-medium model deployed on an NVIDIA Jetson Orin Nano. This Edge AI vision node is seamlessly integrated with an existing IoT dashboard developed at DUT using a lightweight WebSocket protocol. This architecture achieves camera-based real-time inference at the edge while feeding structured alarm events into a centralized React dashboard that actively cross-references temperature and smoke sensor data to dictate threat levels.

III. METHODOLOGY / PROPOSED METHOD

The Intelligent Operations Center (IOC) architecture proposed here aims to address a concrete need: to reliably detect fires and smoke in real time, using multiple data sources. The central idea is to combine two complementary approaches. On one hand, a network of conventional IoT sensors collects physical data as smoke. On the other, an embedded AI-powered vision system analyzes the live video feed. Taken separately, these two approaches each have their weaknesses but combined, they compensate for each other. This section reviews the theoretical foundations of this architecture, then details the vision-based detection strategy, the telemetry protocol used, and finally the decision logic that enables the fusion of information from the different sensors.

A. Overall System Architecture

The system architecture is designed as a decentralized, edge-to-cloud hybrid network. It is fundamentally composed of two main data acquisition branches that converge toward a unified React-based web dashboard (the IOC):

- **Physical Sensing Node:** End-devices (e.g., STM32 and ESP8266/ESP32 microcontrollers) equipped with MQ-2 gas/smoke sensors and temperature sensors continuously monitor the ambient environment.
- **Visual Sensing Node (Edge AI):** An IP camera captures a video stream, which is processed locally by a high-performance edge computing board running a custom

object detection model to identify flames and smoke visually.

Both nodes operate asynchronously. The physical sensors communicate via Wi-Fi, while the Edge AI node transmits its predictions structured as JSON payloads via a local WebSocket server.

B. Vision-Based Fire and Smoke Detection Strategy

Traditional smoke and heat detectors suffer from latency, as they require physical particles or heat to reach the sensor. To mitigate this, computer vision powered by Convolutional Neural Networks (CNNs), specifically the YOLO (You Only Look Once) object detection family, is introduced as an early-warning mechanism.

The strategy relies on a specialized training methodology to handle the complexity of building environments. Because fire emergencies can occur at night, the visual model is trained to process both RGB and infrared (night vision) imagery. This is achieved by intentionally augmenting the training dataset with grayscale images, forcing the neural network to learn structural and morphological features of smoke and flames rather than relying exclusively on colorimetric data. Furthermore, to suppress false positives commonly caused by artificial indoor lighting, the model is trained against negative samples (empty backgrounds and reflections).

C. Real-Time Communication Protocol

To ensure low-latency data transmission between the distributed edge devices and the centralized IOC, a lightweight WebSocket-based communication protocol was chosen over traditional HTTP polling. WebSockets provide a persistent, full-duplex connection ideal for real-time critical alerting.

A significant challenge in decentralized IoT architectures is determining device availability. The IOC dashboard is programmed to flag any node as "Offline" if no packet is received within a 5-second window. Since the visual AI module primarily transmits data upon positive target detection to preserve bandwidth, periods without fire or smoke can trigger false offline statuses. To circumvent this, a "heartbeat" mechanism is theorized: the visual node continuously measures the time elapsed since the last transmission and dispatches a minimal status payload (*awake.json*) at regular intervals if no anomaly is detected.

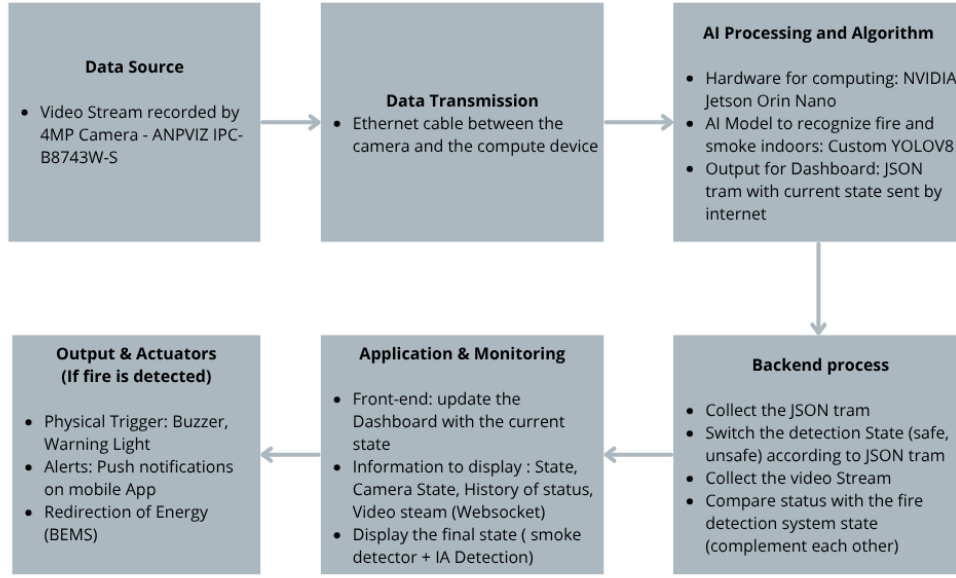


Fig. 1. Overall Architecture of the Proposed Intelligent Operation Center

D. Multimodal Sensor Fusion and Decision Logic

The React-based dashboard acts as the brain of the IOC. Because the physical sensors and the visual AI are developed as independent modules, the dashboard applies a data normalization layer upon receiving incoming JSON payloads to unify varying keys (e.g., standardizing temp, t, and temperature).

To minimize false alarms and optimize emergency response, a deterministic sensor fusion algorithm processes the normalized data. The decision logic is stratified into four strict threat levels:

- **Level 0 (Safe):** Baseline state. No visual anomalies, temperature $< 40^{\circ}\text{C}$, and smoke < 250 ppm.
- **Level 1 (Caution - Early Warning):** Triggered by initial physical anomalies. Temperature $\geq 40^{\circ}\text{C}$ **or** smoke ≥ 250 ppm.
- **Level 2 (Warning - Fire Risk):** Indicates significant environmental deterioration. Triggered when (temperature $\geq 50^{\circ}\text{C}$ **and** smoke ≥ 350 ppm) **or** smoke alone ≥ 500 ppm.
- **Level 3 (Danger - Confirmed Emergency):** The highest threat level, requiring immediate action. Triggered instantly by a visual flame confirmation from the Edge AI (flame == true), **or** extreme physical thresholds (temperature $\geq 60^{\circ}\text{C}$ **or** smoke ≥ 600 ppm).

Upon determining the threat level, the IOC executes automated responses, such as activating UI warnings, physical warning lights (Level 1 and above), and alarm buzzers (Level 2 and above) when operating in auto-mode.

IV. EXPERIMENTAL SETUP / SIMULATION

A. Hardware and Software Configuration

To evaluate the proposed architecture, an experimental testbed was deployed. The Edge AI processing was executed on an **NVIDIA Jetson Orin Nano**, flashed with NVIDIA JetPack 6.2.1 (Ubuntu 20.04). System resources, power draw, and thermal performance during inference were monitored using the *jtop* utility.

The visual input was provided by a 4MP IP camera. Using the SADP network scanning tool, the camera was assigned a static IP address (192.168.0.123), and the Jetson device was configured within the same subnet (192.168.0.100) to ensure a dedicated local connection without relying on an external router. To acquire the video stream efficiently, a custom GStreamer pipeline was utilized to fetch the RTSP stream. By leveraging hardware-accelerated decoding (nv4l2decoder and nvvidconv), the system offloaded the CPU, reserving maximal computational resources for PyTorch inference.

The physical deployment of our Intelligent Operation Center (IOC) edge node is illustrated in Figure 2. The setup comprises the NVIDIA Jetson Orin Nano as the core processing unit, interfaced with the camera for real-time video capture. This configuration allows us to evaluate the model's inference capabilities within the spatial and thermal constraints of a real-world edge deployment.

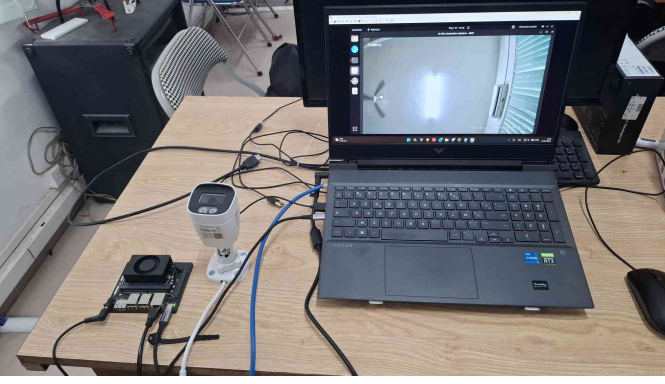


Fig. 2. Complete hardware setup of the Edge AI node, featuring the NVIDIA Jetson Orin Nano and the camera.

B. Dataset Description and Model Training

The object detection model was developed iteratively across four distinct phases to maximize precision and eliminate false positives, transitioning from YOLOv8-small (YOLOv8s) to the more capable YOLO11-medium (YOLO11m) architecture:

- 1) **Phase 1 (Prototype):** A baseline YOLOv8s model was trained on 13,700 images (8,000 fire instances, 9,000 smoke instances). To simulate infrared night vision, 22% of the dataset was converted to grayscale. However, the lack of negative samples caused hallucinations in complex environments especially when neon light appears.
- 2) **Phase 2 (Background Addition):** The architecture was upgraded to YOLO11m, and 1,302 background images (empty rooms) were added, totaling 15,500 images. This improved training stability.
- 3) **Phase 3 (Data Augmentation):** Extensive augmentations (rotation $\pm 10^\circ$, brightness $\pm 25\%$, and 2px blur) were applied, expanding the dataset to 24,000 images and increasing the target instances to over 13,000 for fire and 14,700 for smoke.
- 4) **Phase 4 (Final Refinement):** To combat specific false positives caused by neon lights and window reflections, 150 highly specific background images were injected alongside 1,050 new fire images. Photometric augmentations (HSV alterations, flip horizontal, mosaic) were pushed further.

The final Phase 4 dataset comprised 26,000 images. Training was executed using the Stochastic Gradient Descent (SGD) optimizer (learning rate = 0.01, momentum = 0.937, weight decay = 0.0005) over 100 epochs with a batch size of 8.

To achieve real-time inference (25 FPS) on the edge hardware, the final trained PyTorch model (.pt) was exported to the NVIDIA TensorRT format (.engine) utilizing FP16 precision. The confidence thresholds were empirically set to 0.4 for general prediction and 0.6 to trigger a confirmed detection alert.

C. Network Implementation

The telemetry system was implemented using a custom local WebSocket broker (serveur_relais.py) hosted on port 8765, acting as a bridge between the Jetson inference

script and the React IOC dashboard. To comply with the dashboard's 5-second timeout restriction, the vision script (fire-detect.py) was programmed with a SEND_DELAY of 3.0 seconds. If 3 seconds elapse without a fire or smoke detection, the system automatically transmits an awake.json packet indicating nominal status, thereby ensuring uninterrupted real-time tracking without saturating the local network.

V. RESULTS AND DISCUSSION

A. AI Model Evaluation and Comparison

To validate the efficiency of our visual fire detection module, we compared our initial baseline prototype (V1), based on the YOLOv8s architecture, with our final optimized model (V4) using the latest YOLO11m architecture. As detailed in Section IV, this final version is the culmination of an iterative process involving extensive dataset balancing, advanced hyperparameter tuning (such as HSV augmentation and mosaic scaling), and architectural upgrades. The training process for this final model was conducted over 100 epochs with an image size of 640x640 and a batch size of 8. The complete configuration of the training parameters is detailed in Table II.

TABLE II
YOLO11M TRAINING PARAMETER CONFIGURATION

Parameter	Value
Model Architecture	YOLO11m
Epochs	100
Image Size (imgsz)	640
Batch Size	8
Optimizer	SGD
Learning Rate (lr0)	0.01
Momentum	0.937
Weight Decay	0.0005

The initial YOLOv8s model demonstrated promising baseline metrics but suffered from a critical vulnerability in real-world indoor environments: it frequently misclassified artificial light sources, such as neon lights and reflections, as fire. Analysis of the V1 dataset revealed a lack of negative examples (empty background images). To address this, the dataset was massively expanded for the V4 training, incorporating complex indoor backgrounds, resulting in a robust dataset of 12,490 fire instances and 14,100 smoke instances.

TABLE III
PERFORMANCE COMPARISON BETWEEN YOLOv8s (V1) AND YOLO11m (V4)

Metric	YOLOv8s (V1)	YOLO11m (V4)
mAP50	~ 84.0%	89.5%
False Positives (Neon/Lights)	High vulnerability	Mitigated
Robustness to complex backgrounds	Low	High

As illustrated in Table III and Figure 3, the transition to YOLO11m coupled with the enhanced dataset significantly improved the model's performance. The final mean Average Precision (mAP50) stabilized at 89.5%. Furthermore, the confusion matrix (Figure 4) confirms this high accuracy, with 862 true positive fire detections and 956 true positive smoke detections. The remaining false positives (151 instances of background classified as fire) represent a drastically reduced

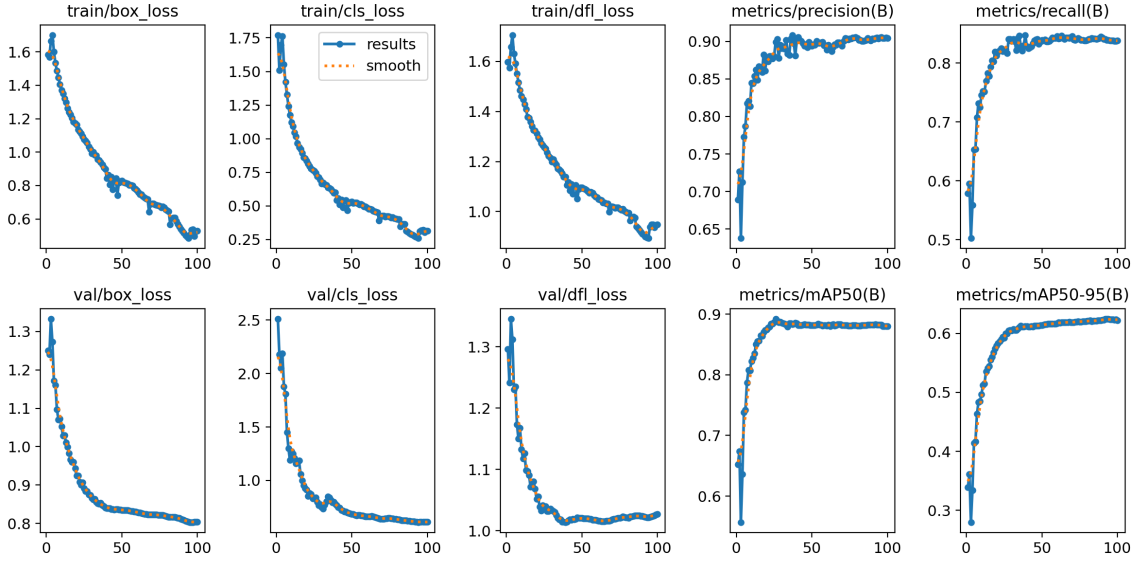


Fig. 3. Training and validation metrics for the final YOLO11m model.

False Positive Rate considering the massive dataset expansion, proving the model's acquired robustness against indoor lighting anomalies.

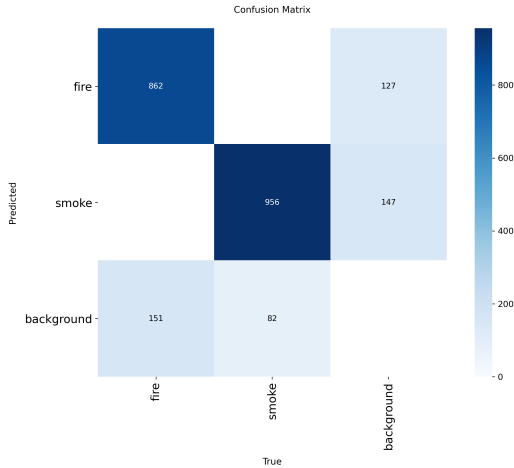


Fig. 4. Confusion matrix of the YOLO11m model showing high true positive rates.

The qualitative improvement of the model in real-world scenarios is particularly noteworthy. During the initial testing phase with the V1 model, artificial indoor lighting, specifically neon tubes and bright reflections on reflective surfaces, consistently triggered false positive fire alerts. This is a common limitation in vision-based fire detection systems that rely solely on color or basic shape feature extraction. By curating a targeted negative dataset (images containing complex lighting but no fire) and leveraging the advanced feature extraction capabilities of the YOLO11m backbone, the model learned to contextualize light sources. The drastic reduction in background-to-fire misclassifications demonstrates the model's enhanced spatial awareness, making it highly suitable for indoor smart building

environments where artificial lighting is ubiquitous.

B. Edge AI and IoT Dashboard Integration

The trained YOLO11m model was deployed on the NVIDIA Jetson Orin Nano, acting as an Edge AI node. To eliminate the high latency associated with cloud databases (such as Firebase), we developed a custom local WebSocket relay server. This allows the Jetson to transmit visual inference data (Fire/Smoke detected) in real-time to a centralized React-based dashboard.

Traditional IoT architectures often rely on REST APIs and polling mechanisms, which introduce significant latency and network overhead, rendering them unsuitable for critical emergency responses. By implementing a full-duplex WebSocket connection, the Jetson Orin Nano maintains a persistent, low-latency communication channel with the React dashboard. This ensures that the moment a bounding box confidence threshold exceeds our predefined limit, the event is immediately pushed to the client interface.

To maintain network stability, a 4-second JSON "Heartbeat" (ping) was implemented, ensuring the dashboard actively monitors the camera's online status. The system utilizes a multi-tier alert logic ranging from Level 0 ("AN TOÀN" / Safe) to Level 3 ("NGUY HIỂM" / Danger). During our real-world tests, the integration of visual data with the physical sensor telemetry (temperature and MQ-2 gas detectors) proved highly effective. If the camera detects a potential flame, the system cross-references this visual alert with the physical sensors. This sensor-fusion approach ensures that isolated visual false positives do not trigger a general building alarm unless corroborated by physical environmental changes. Figure 5 demonstrates the dashboard successfully registering a Level 3 threat during a controlled fire test.

In addition to the quantitative training metrics, Figure 6 showcases a qualitative result of the system in action. The YOLO11m model accurately draws a bounding box around

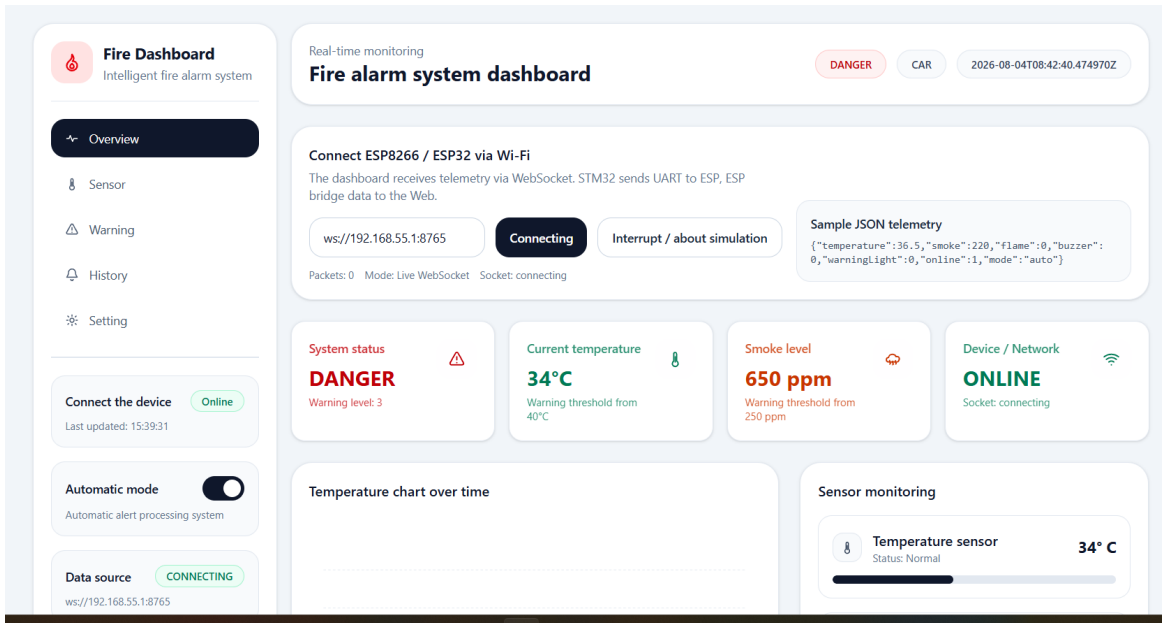


Fig. 5. Real-time React Dashboard triggering a Level 3 (Danger) alert during live testing.

a live fire during our controlled real-world experiments. This demonstrates high confidence and precise spatial localization capabilities at the edge, which directly triggers the Level 3 alert on the centralized dashboard.

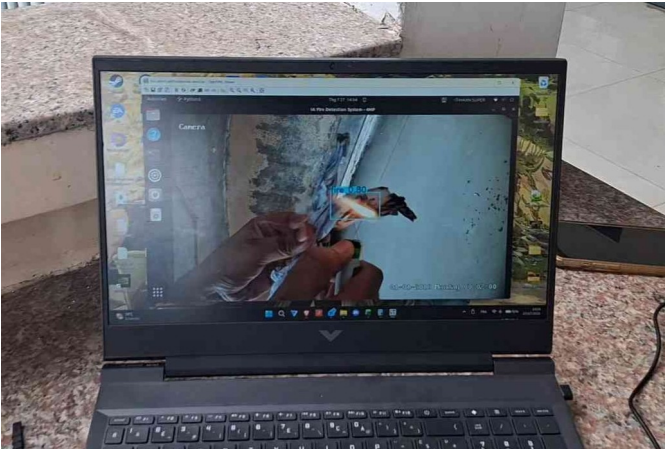


Fig. 6. Real-time qualitative inference: YOLO11m successfully detecting and bounding a real fire in an indoor environment.

C. Edge Hardware Performance

Deploying deep learning models on Edge devices requires strict hardware optimization to ensure real-time performance without thermal throttling. To achieve this, the YOLO11m PyTorch model (.pt) was converted into a TensorRT engine (.engine). During this compilation, we enabled half-precision floating-point format (half=True / FP16). This optimization significantly reduced the memory footprint and computational load.

Standard deep learning models operate in 32-bit floating-point precision (FP32). While highly accurate, FP32 operations are computationally expensive and consume substantial

memory bandwidth. By quantizing the YOLO11m model to FP16 via TensorRT, we effectively halved the memory requirements for the network's weights and activations. This quantization process exploits the specialized Tensor Cores available on the Jetson Orin Nano's Ampere architecture, accelerating matrix multiplication operations with minimal loss in inference accuracy. This synergy between software optimization (TensorRT) and hardware architecture (Tensor Cores) is what ultimately enables the system to process high-resolution video streams locally.

Furthermore, to prevent the "System throttled due to over-current" hardware protection from triggering, the inference frame rate was reduced to 12 frames per second, instead of the 24 frames per second natively provided by the camera. This reduction in processing load helped limit the module's power consumption and avoid the current spikes responsible for triggering this protection.

As monitored via the `top` utility (Figure 7), the system demonstrated exceptional stability during live inference. While processing the video feed and running the YOLO11m TensorRT engine, the Jetson maintained a smooth real-time processing speed of approximately 12 FPS. The hardware utilization remained highly efficient, with the GPU usage dynamically fluctuating based on frame complexity but averaging around 30% to 40% capacity, and the RAM usage peaking at 2.7 GB. The overall power consumption of the board remained exceptionally low and stable around 8.1 W. These metrics prove that our proposed Edge AI system is highly optimized and perfectly capable of running continuous 24/7 surveillance in a smart building without overheating.

VI. CONCLUSION AND FUTURE WORK

In this paper, we proposed and successfully deployed a hybrid Intelligent Operation Center (IOC) for building safety, synergizing Edge AI computer vision with Internet of Things

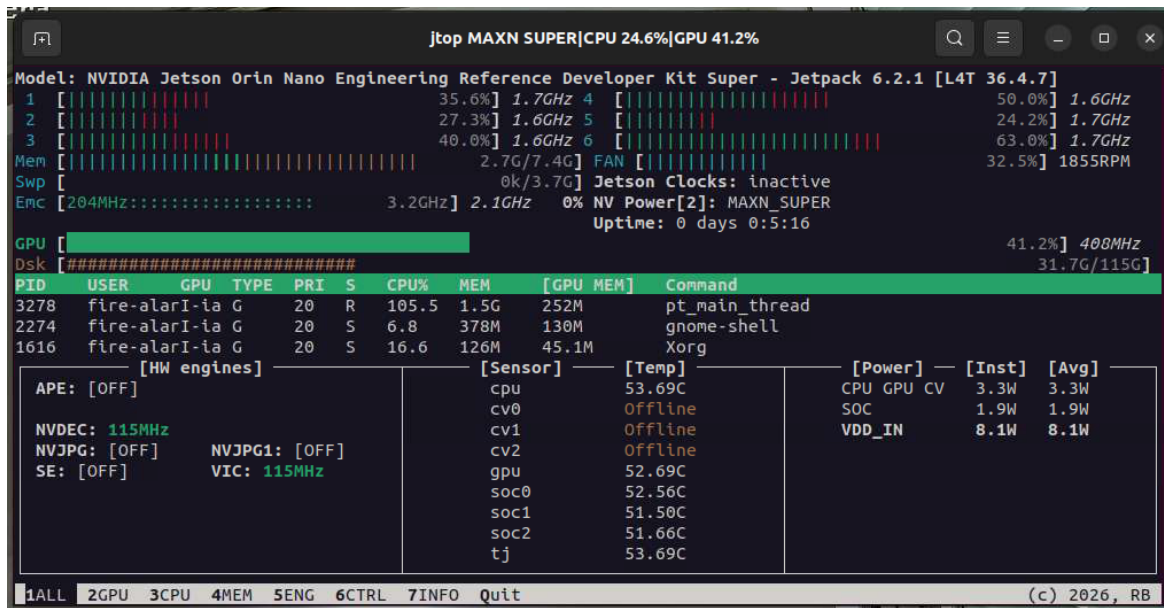


Fig. 7. System resources monitoring during live inference using jtop.

(IoT) principles. By transitioning from a baseline YOLOv8s architecture to a highly optimized YOLO11m model, and by massively expanding our dataset to include complex indoor backgrounds, we effectively mitigated a major limitation of vision-based fire detection: false positives triggered by artificial lighting and neon reflections. The final visual model achieved a robust 89.5% mAP50.

Furthermore, the deployment of this model on an NVIDIA Jetson Orin Nano demonstrated the viability of Edge computing for critical emergencies. Through TensorRT FP16 quantization, the system achieves real-time inference without thermal throttling, maintaining an exceptionally low power footprint even in unrestricted performance modes. By replacing high-latency cloud databases with a custom WebSocket relay server, the visual data is seamlessly fused with simulated sensor telemetry in a real-time React dashboard, enabling instantaneous threat level evaluation.

While the current system proves highly effective, several avenues for future work remain to enhance its scalability and integration:

- **Complete Physical Sensor Integration:** Finalizing the hardware connections to incorporate live data from physical MQ-2 gas detectors and temperature sensors (e.g., via ESP32 microcontrollers) to fully realize the multimodal sensor-fusion architecture.
- **Cloud Logging and Analytics:** Reintegrating a cloud database system, such as Firebase, strictly for asynchronous logging and historical data analysis, without impacting the low-latency local alert system.
- **Multi-Camera Scalability:** Evaluating the Jetson Orin Nano's computational capacity to handle multiple concurrent video streams, enabling comprehensive coverage of larger building complexes.
- **Mobile Alerting System:** Developing a dedicated mobile application or integrating a messaging bot (e.g., Tele-

gram) to push real-time notifications and image snapshots directly to the smartphones of security personnel and facility managers.

Ultimately, this research demonstrates that combining optimized deep learning models with edge-based IoT dashboards provides a highly reliable, low-latency, and scalable solution for modern smart building fire safety.

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